



CERTIFICATE OF ANALYSIS

HESS CORPORATION
10340 68th St. NW, Tioga, North Dakota,
USA-58852

V-TIC SERVICES

VTIC LAB NUMBER	W1075	FILE NUMBER	ND231079		
LOCATION	Blue Buttes Compressor Station	SOURCE OF SAMPLE	Inlet Natural Gas Line Sample Valve		
PRODUCT TYPE	Natural Gas	SAMPLE(s) SUBMITTED BY	Troy Cundiff		
GRADE/SPECIFICATION		CLIENT REFERENCE			
SEAL NO	308				
BBCS Inlet (Spot) - W1075.003					
SAMPLED DATE & TIME: 03-Mar-2023 1320 Hrs / RECEIVED DATE & TIME: 03-Mar-2023 1900 Hrs / DATE TESTED: 22-Mar-2023					
Method Number	Method Name	Min	Max	Test Results	Unit
GPA 2286	Extended Analysis for Natural Gas & Similar Gaseous Mixtures by Temperature Program Gas Chromatography				
	Full Analysis	-	-	See Attached	-

Comment: Pressure = 21 psi
Temp = 32°F
H2S = 5 ppm
Cylinder# 308

AUTHORISED SIGNATORY

Curtis Kendal
Branch Manager
100000000000000454
Date: 22-Mar-2023 1612 Hrs

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V-TIC Services Inc_Williston, N.D.

C10+ Extended Natural Gas Analysis Report

Sample Information

	Sample Information
Lab	V-TIC Services Inc_Williston, N.D.
Analysis Number	W1075.003
Customer	Hess Monthly (Donny)
Sample Name	BBCS Inlet Natural Gas Line
Field	
Property	BBCS
Well ID	
Meter	
Sample Point	Sample Valve
Sample Type	Natural Gas
Cylinder Number	#308
Pressure PSIG	21 psi
Temp F	32°F
Field H2S ppm	5 ppm
Moisture (mmcf)	lbs/MMCF
Sample Technician	Troy Cundiff
Sample Date	3/3/23 @ 13:20pm
Lab Technician	Sean Weiner
Reviewed by	Curtis Kendal
Comments	File#ND231079
Methods/Calculations/Constants	-----GPA 2286, GPA 2172, GPA 2145-----
Report Date	2023-03-22 14:00:33

Component Results

Component Name	Norm Mole%	Norm Weight%	Norm Volume%	GPM (Dry) (Gal. / 1000 cu.ft.)
Nitrogen	2.325	2.559	1.221	0.257
Methane	60.042	37.849	48.602	10.231
Carbon Dioxide	1.106	1.913	0.901	0.190
Ethane	19.726	23.307	25.188	5.302
Propane	9.954	17.247	13.093	2.756
i-Butane	1.141	2.606	1.783	0.375
n-Butane	3.507	8.010	5.279	1.111
i-Pentane	0.710	2.013	1.240	0.261
n-Pentane	1.092	3.096	1.890	0.398
n-Hexane	0.309	1.046	0.606	0.128
n-Heptane	0.075	0.295	0.165	0.035
n-Octane	0.012	0.054	0.029	0.006
n-Nonane	0.001	0.005	0.003	0.001
n-Decane	0.000	0.000	0.000	0.000
Total:	100.000	100.000	100.000	21.051

Results Summary

Result	Dry	C6+
Gross Heating Value (BTU / Ideal cu.ft.)	1452.7	4961.4
Gross Heating Value (BTU / Real cu.ft.)	1460.4	
Gross Heating Value (BTU / lbm)	21612	20913
Gross Heating Value (BTU / gal.)	69375	116965
Relative Density (G), Ideal	0.8787	3.1012
Relative Density (G), Real	0.8830	

Result	Dry	C6+	
Compressibility (Z) Factor	0.9947		
Total Molecular Weight	25.449	89.820	
Total Vapor Pressure (psia)	3311.51	4.18	
Total Volume (cu.ft. / gal.)	47.7578	23.5755	

Total Component Results

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